MIL-C-23334B(Wep)
27 October 1964
Superseding
MIL-C-23334A(Wep)
7 April 1964

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 33 MOD 1

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 33 MOD 1 impulse cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
NAVWEPS OS 9862	Propellant - Double Base (Nitrocellulose-Nitroglycerine)
NAVWEPS OS 12877	Project Charge Practice, MARK 12 MOD 0

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

FSC 1375

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DRAWINGS

Bureau of Naval Weapons

LD 537914	Cartridge, Impulse, MARK 33 MOD 1
LD 537915	Container, Inner, for Cartridge, Impulse, MARK 33 Type
LD 537916	Container, Outer, for Cartridge, Impulse, MARK 33 Type
LD 546641	Element, Ignition, MARK 10 MOD 0
LD 161978	Projector Charge, Practice, MARK 12 MOD 0 Assembly
Dwg. SA 2640784	Projector, Single Spigot

PUBLICATIONS

SECNAV Instruction 4030.3B	Uniform Preservation, Packaging, Packing and Marking of Items of Military Supply

Bureau of Naval Weapons

OP 5	Ammunition Ashore (This publication is for the exclusive use of the Government inspector. It is not to be furnished to bidders or contractors.)
NAVORD Instruction 8010.24	Marking of Federal Stock Numbers and Nomenclature on Ammunition

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

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CODE OF FEDERAL REGULATIONS

49 CFR 71-90	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 33 MOD 1 impulse cartridge covered by this specification shall be manufactured in accordance with LD 537914 and all documents listed thereon.

3.2 Material. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective BUWEPS drawings unless specific approval in writing covering a departure therefrom has been obtained from the Bureau of Naval Weapons prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 84 MARK 33 MOD 1 impulse cartridges shall be furnished the Government prior to starting on full production. See 4.3.2.

3.3 Primary components. For the purposes of this specification the MARK 10 MOD 0 ignition element is defined as a primary component.

3.4 Ignition element. The MARK 10 MOD 0 ignition element shall conform, in all respects, to BUWEPS LD 546641.

3.5 Propellant charge. The propellant used in the MARK 33 MOD 1 impulse cartridge shall be manufactured in accordance with, and meet the requirements of, the Type 1 propellant of NAVWEPS OS 9862.

3.5.1 Quantity. All propellant charges shall be as specified on Bureau of Naval Weapons drawing 2240758. Charge shall be determined by weight rather than by volume measurement.

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3.5.2 Smokeless powder. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. Inspect for signs of decomposition as outlined in paragraphs 1308-1318 and 1322 of OP 5. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.6 Cartridge inspection.

3.6.1 Visual inspection. The complete cartridge shall be free of the visible defects noted in 4.7.1.

3.6.2 Electrical resistance. The resistance of the bridges in each completely assembled cartridge manufactured, when measured electrically as specified in 4.7.2 shall be not less than 0.08 ohm or more than 0.15 ohm.

3.7 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.3, shall meet the requirements listed below when subjected to the test prescribed in 4.7.3.

3.7.1 Range. The upper (U) and lower (L) limits of the range shall be 1000 and 700 feet respectively. See 4.7.3.1.1.

3.7.1.1 Misfire. There shall be no misfires, see 4.7.3.1.2.

3.7.1.2 Hangfire. There shall be no hangfires, see 4.7.3.1.3.

3.8 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

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4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-414.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 84 cartridges shall be prepared. A preproduction sample acceptable for functional and environmental testing shall be defined as a sample which has met the requirements of paragraphs 3.4 through 3.6.2 of this specification. The contractor shall deliver the cartridges to the activity designated by the contracting agency for test. These cartridges shall be manufactured as one sample. Preproduction sample acceptance shall consist of the following tests as prescribed in Table 3 of MIL-D-21625: 40-foot drop (6), 15 g shock (12), temperature and humidity cycling (18), vibration (12), +70°F (12), +160°F (12) and -20°F (12) which shall be used in lieu of the -65°F listed in Table 3 of MIL-D-21625. Test results shall fall within the limits set forth in 3.7.1 through 3.7.1.2. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction cartridges will not be applied as a part of the quantity specified for delivery by the contract.

4.3.3 Production lot. The suggested production lot size is 1301 minimum and 8000 maximum cartridges. Cartridges required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic test shall be delivered to the activity designated by the contracting agency.

4.4 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Production and acceptance inspection. Production and acceptance inspection shall consist of the following:

Item	Nature of Test	Requirement	Test
Charge	Quantity	3.5.1	- - -
Smokeless powder	Condition	3.5.2	- - -
Cartridge	Visual inspection	3.6.1	4.7.1
Cartridge	Electrical resistance	3.6.2	4.7.2
Cartridge	Ballistic	3.7	4.7.3

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4.6 Packing, packaging and marking. The inspector shall ascertain that the packing and packaging of the cartridges and their markings conform to Section 5 of this specification.

4.7 Tests.

4.7.1 Visual inspection. Examine each cartridge produced under a contract and reject for dents, deep scratches, split or cracked lip in the crimped area, perforated closure discs, sharp edges, burrs, presence of sealant on exposed surface of ignition element electrode, and other defects which may conceivably prevent entry into the firing chamber or affect stability or resistance to moisture.

4.7.2 Electrical resistance. Conformance with the bridge circuit resistance of 3.6.2 shall be ascertained by means of a test circuit which limits the bridge current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of the current. A suitable bridge, which is known to be accurate within 0.005 ohm in the range of resistance specified, shall be used to determine whether or not the requirement is being met. Units which conform to the requirements of 3.6.2 shall be acceptable.

4.7.3 Ballistic test.

4.7.3.1 The number of cartridges selected from each lot for the ballistic test shall be in accordance with the requirements of Table A2 of MIL-STD-414, Inspection Level IV and an AQL of 1.0. The cartridges shall be conditioned for 24 hours at +70°F ± 5°F and fired in a MARK 12 MOD 0 practice projector charge which shall be mounted on a single spigot projector constructed in accordance with BUWEPS Dwg. SA 2640784 or equivalent. The MARK 12 MOD 0 practice projector charge used for this test shall conform in all respects to BUWEPS LD 161978 and OS 12877. The spigot shall be oriented so that its angle of projection is 55° from the horizontal and 0° mount train order. The firing current shall be 6 volts dc or 20 volts ac with a minimum of 5 amps. The criteria of acceptability shall be based on the following:

4.7.3.1.1 Range: The lot shall be acceptable if the total estimated percent defective in the lot ($P = P_U + P_L$) is less than the maximum allowable percent defective M where U and L are 1000 and 700 feet respectively. The estimated percent defective shall be computed in accordance with Section B, Part II of MIL-STD-414.

4.7.3.1.2 Misfire. One or more misfires shall cause rejection of the lot unless the misfire is clearly attributable to faulty test equipment or procedure.

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4.7.3.1.3 Hangfire. One or more hangfires shall cause rejection of the lot. A hangfire is obtained if the interval between actuation of the firing mechanism and first indication of MARK 12 MOD 0 practice projector charge movement is greater than 0.5 second.

4.8 Retest. There shall be no retests.

4.8.1 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and the test reconducted with a new sample of cartridges.

5. PREPARATION FOR DELIVERY

5.1 Packaging. In addition to the following, Level A and Level C packaging shall be in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. For a definition of Level A packaging see 6.4.1.

5.1.1.1 Inner container. Four cartridges shall be packaged in a ccordance with LD 537915.

5.1.2 Level C. Minimum packaging for those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be Level C. For a definition of Level C packaging see 6.4.1.1.

5.2 Packing. In addition to the following, Level A and Level C packing shall be in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. For a definition of Level A packing see 6.4.2.

5.2.1.1 Outer container. One hundred inner containers as described in 5.1.1.1 shall be packed in accordance with LD 537916.

5.2.2 Level C. Minimum packing for those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be Level C. For a definition of Level C packing see 6.4.2.1.

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5.3 Marking. Cartridge marking shall be as specified on Bureau of Naval Weapons Drawing 2240758 and NAVORD Instruction 8010.24. In addition, packaging and packing markings shall be in accordance with the explosive section of the Code of Federal Regulations 49 CFR 71-90, MIL-STD-129 and NAVORD Instruction 8010.24.

6. NOTES

6.1 Intended use. The MARK 33 MOD 1 impulse cartridge is intended for use in the MARK 12 MOD 0 practice projector charge.

6.2 Ordering information. Procurement documents should specify the following:

a. title, number, and date of this specification;

b. whether preproduction inspection is required and if so specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NavWeps Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Bureau of Naval Weapons and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definitions of Levels A and C packaging and packing given below were extracted from SECNAV Instruction 4030.3B.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment, handling, indeterminate storage and world-wide redistribution.

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6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity, for immediate use or controlled humidity storage. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations, and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 General safety precautions.

6.5.1 The loading, assembly, and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosives precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer (loader) of the above items should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail provisions of the specifications.

6.5.2 All loading operations should be conducted in a neat and orderly manner.

6.5.3 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

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6.5.4 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.5 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.6 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosive Safety Board covering quantity-distance regulations for explosives, shall apply.

6.5.7 Proper care should be exercised at all times to protect personnel from accidents, fires or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed:

6.5.7.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.7.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.7.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.5.7.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment and fixtures; and, where necessary, employ suitable grounded conductive covering for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to

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initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heating apparatus, etc. Where necessary for safety, humidity (see para. 6.5.5) of work rooms should be increased, as required to lessen electrostatic effects without excessive moisture absorption.

6.5.7.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc. by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric types of primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.5.7.6 Enforce, where necessary, the wearing or suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.7.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.7.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-23334B(WP) CARTRIDGE, IMPULSE, MK 33 MOD 1

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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